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Course Content

Topics
Resume , ATS Score and CV preparation
Whats the current market scenario and what to prepare.
Way to prepare.
C# OOP , Threads and SOLID
Delegates and Events
Memory Management :- Stack, Heap, Value types, Reference types, Boxing, Unboxing and Garbage collector
OOP, interfaces, abstract classes.
SOLID , DI and IOC
Generic and Collections (Concurrent) , IQueryable , IEnumerable and Yield.
Multi-threading , TPL , Async and Await
LINQ Questions
ASP.NET
Middle wares and Filters
Session and Cookies
DI Types in ASP.NET
Authentication and Authorization and JWT Token
ASP.NET WebAPI
EF and ORM
SQL Server



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Normalization
CTE Questions
Get top N Query
Delete duplicate
Architecture , Design pattern ,DDD
Covering 4 important Design patterns
Microservices and DDD interview preparation
Azure
Interview prep around App services , Function App , Logic Apps
Angular interview preparation
Performance Questions
Resume prep , linked
Mock interview and Mock interviews

Do not give shallow academic answers or non-fruitful by product answers.

Display Authority by displaying them the real use of concept.

Technical Vocabulary.

Beating around bush seniors do not time, what happens at back end , they do not get salary.

Silly questions do not make mistakes , do not hurry. Trust...

Interview preparation section.

Delegates and Events

1. Answering Delegate is pointer to a function, Answering it is type safe , technically right but logically you cannot convince the interviewer and neither get convinced.
2. Delegate acts like abstraction (different behaviours) but that can be nicely done by Interfaces.
3. Academic definition and By product...
4. Delegates are callbacks. (Displaying Authority)
5. Multicast delegate +=
6. Delegates are not protected are not encapsulated.
7. Block and Nonblock words please use in interviews.
8. Event is a encapsulation over delegates , they use delegates internally.
9. Event helps to implement PURE Publisher Subscriber , broadcast (Technical vocabulary)
10. Delegate is not used too much but Events (Display authority , practical ness).
11. Errors , Examples of Events...



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12. Built in delegates :- Flavors of Delegates. , Action , Func , Predicate and Lambda, Arrow function(Syntactic sugar over delegates) ,

Memory Management

- Background process which run un determentically and clean unused **managed** objects.
- GC does not claim unmanaged objects.
- Managed :- CLR , Unmanaged :- Non.NET
- Dispose pattern , unmanagaed. GC makes a double trip for destructors.
- GC 0 ,1 and 2 – How old the object are in memory
- Value ar allocated on stack while ref are allocated on a heap.
- Value – ref (boxing) , ref – v (Unboxing)
- Stack will have the memort pointer.
- Not completely but uncessarily do it or else we have performance implication.

OOP Principles

- 4 pillars :- Abstraction , Poliymorhism , Inheritance and Encapsulation. (APIE)
- Abstraction :- Most needd (essential) properties and methods. Its more the planning phase but do nt use the word planning.
- Abstraction in code is represented by interfaces.
- Wrong Abstraction :- Interfaces (LISKOV)
- Interfaces :- Conarct , Implemented , Enforced , represent abstraction.
- We will also walk through real time questions listed below.
- Inheritance : Patyen child relations , where child etting
- You can only override if you have virtual.
- Polymorphsim :- Poly means many morp change as per situation , example Employee parent object can point towatds its child object like MangerEmployee , HR Employee Dynamics Polymorphism
- Static polymoprhsim :- Method overloading. Same method can ve called with different signatuire.
- Abstract classes :-
- Abstract class IS NOT Abstraction (interfaces).
- Interfcae vs Abstract classes :- They are different. Interface is abstraction while abstract class is a half defined parent.
- What if I make all methods as abstract , technically it's a interface and it should be made an interfgeace rather than absyrat class.
- Empty abstract class was put for placeholder ahead for logic Is not needed , cureen c# Default interface solves it.
- Shadowing you are saying these methods will be onvoked from that class instance in case of polymorphism these methods will not work.
- Loosely coupled code :- Interfaces (Abstraction) + Polyphorism
- You CAN NOT create instance of Interface , Abstract classes



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- Implicit and Explicit interfaces.
- Partial classes and when to use.

SOLID Principles , DI and IOC

- SRP and OCP are easy.
- LISKOV
- DI is dependency inversion
- ISP

Task and Threads

- Synch processing :- One after another. Blockin calls.
- Callers intention is Multi threading , Parallelism.
- Task represents an Async (Concurrent) operation.
- For parallel execution we use threads and for Async we use task which provides the callback in a promise/future.
- Definition In C#, a **thread** is the smallest unit of execution within a process. It represents an independent path of execution, allowing your program to perform multiple tasks simultaneously.
- Process and threads.
- How to debug threads , Debug , Window show threads.
- Main thread and Worker thread.
- In threads , Caller created the thread expclicitly , in async in purest form (Not CPU Bound) should not have threads.
- Task vs Threads
- Async and Await helps to wait when the task finishes
- Questions on Asynchronous programming, threads, multitasking, TPL, Parallel processing
- Run → Safe Thread Pool /Start → Manual + current scheduler /Start New → Advanced control
- State machine in Asynch and Await.

Collections and Generic collections

- Array is of same data type (Strongly typed) and its of fixed length. ArrayList is dynamic and has flexible data type. Hashtable allows to you look up using your own key.
- ArrayList and Hashtable do boxing and unboxing.
- Generic collection are strongly type and flexible , List<>
- Generics :- Decouples your Data type and Logic. When you are giving example repository is one of the best examples.
-



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Architecture expression

- C4 Modelling, Strategic DDD
- UML class , Sequence diagrams

LINQ Preparation

Post vs Put vs Delete vs Patch vs Get

- Records , Post - insert , Put - update....
- Post insert and Put is for update.
- Do not use records :- Resource(Record , Text file , Blob).
- Use the word Idempotent.
- In put new resource can get added and updated , but the identifier remains same.

Entity Framework

- Home work

Pagination questions

SELECT * FROM Products

ORDER BY ProductID

OFFSET (@PageNumber - 1) * @PageSize ROWS

FETCH NEXT @PageSize ROWS ONLY;

From EF Skip and Take

Key set pagination.

SELECT TOP (@PageSize) * FROM Products

WHERE ProductID > @LastSeenID

ORDER BY ProductID;

Indexes on order by clause.



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How will you design ZOMATO kind application ?

HASCCSTQO

1. High-Level Architecture

Explain that the application would follow a **Microservices Architecture** to ensure independent scaling of different business functions.

- **Front-end:** Mobile apps (Xamarin/MAUI or Flutter) and Web apps (ASP.NET Core MVC or Blazor).
- **API Gateway:** A centralized entry point using YARP or Ocelot to handle authentication (OAuth 2.0/JWT), rate limiting, and request routing.

2. Core Microservices and Features

Break down the system into logical services, each with its own database if necessary to maintain loose coupling.

- **User Service:** Manages customer profiles and authentication using ASP.NET Core Identity with JWT tokens for secure sessions.
- **Restaurant Service:** Handles restaurant onboarding, menu management, and searches using Elasticsearch for fast, geo-spatial queries (e.g., "Indian food pav bhaji").
- **Order Service:** Manages the order lifecycle from creation to completion. It uses the Outbox Pattern to ensure data consistency across services.
- **Payment Service:** Integrates with gateways like Stripe or Razorpay, using transactional processing to ensure atomic order confirmation and payment.
- **Delivery Service:** Assigns the nearest delivery partner using location-based logic and provides real-time GPS tracking.

3. Key Technical Implementations

- **Real-time Updates:** Use SignalR to push live order status updates (e.g., "Restaurant accepted," "Out for delivery") to users without refreshing the app.
- **Asynchronous Processing:** Use Apache Kafka or Azure Service Bus to decouple services. For example, when an order is placed, a message is published so the Payment and Notification services can act independently.
- **Caching:** Use Redis for fast session lookups and to store frequently accessed data like popular restaurant menus to reduce database load.
- **Data Storage:** Use a mix of SQL Server or PostgreSQL for transactional data (Users, Orders) and NoSQL like MongoDB for unstructured data like user reviews.



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4. Database Design (Relational)

Mention key tables and their relationships:

- **Users:** UserID, Name, Address, Role (Customer, Admin, Partner).
- **Restaurants:** RestaurantID, Name, CuisineType, Location (Lat/Long).
- **Menu Items:** ItemID, RestaurantID, Price, Availability.
- **Orders:** OrderID, UserID, RestaurantID, TotalAmount, Status.

5. Performance & Scalability

- **Horizontal Scaling:** Explain how you would deploy services in Docker containers orchestrated by Kubernetes (K8s) to handle traffic spikes during peak meal hours.
- **Monitoring:** Use OpenTelemetry for distributed tracing and Prometheus/Grafana for monitoring system health.

Coding Questions

1. To not use in-built functions and should have good performance.
2. Write on Notepad → Browser → VSCode / Visual Studio.
3. Camel Case and Pascal Case , namespaces.
4. Berris MasterMind
VeriKlick
TestDom
AI TalentQuest (WebMobi 360)
Zinterview.ai
5. Performance is always measured by BIG O notation.
6. String is an Array in C# with 0 Index.
7. String are immutable.
8. L shape LESS THAN. Keep one in Mind.
9. Writing the Forloop back and forth.
10. Understanding of collections very important.



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Real time Questions

OOP / SOLID

- 1) What is method signature?
- 2) What is the Order of constructor execution?
- 3) [What is destructor and how many can be there in a class\(only 1\)?](#)
- 4) [Why do you need a destructor when GC is there ?](#)
- 5) What happens when constructor is private?
- 6) If an Abstract class has concrete method so can its child class have same method name and can it modify it?
- 7) Abstract vs Static class
- 8) [Abstract vs Interface](#)
- 9) Why abstract class has constructors
- 10) When to use abstract and when to use interface give real time example in c#
- 11) Implicit and explicit interface
- 12) What is default interface and why do we need it ?
- 13) Polymorphism in c#
- 14) Method overriding with example
- 15) Difference b/w Virtual, Override and new keywords in C# ?
- 16) Method hiding with example
- 17) Is return type considered in method overloading?
- 18) Why and when to use method hiding? give real time scenario
- 19) How to extend a sealed class?
- 20) What is extension method? Write an example
- 21) What is solid principle and why to implement?
- 22) State all the principles with example
- 23) What is Liskovs Substitute Principle, explain with example
- 24) Difference in Open close and Liskovs principle
- 25) Diff between Liskov vs ISP ?
- 26) Calculate salary of employee based on grades (strategy pattern)



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C#

THEORY QUESTIONS -

- 1) Value type vs Ref type.
- 2) Ref vs Out, give real time example and usage.
- 3) Can ref keyword be used in Async programming?
- 4) Const vs Readonly.
- 5) What is garbage collector?
- 6) How to remove unmanaged code?(How we remove garbage in unmanaged code?)
- 7) Throw vs Throw ex.
- 8) What are access modifiers?
- 9) What is Partial class give real time scenario.
- 10) What are generics? Have you implemented any in your project?
- 1) What are collections and its type.
- 2) What is Dictionary, List, HashTable.
- 3) What is Stack and Queue?
- 4) Difference between Generics and non generics?
- 5) What is Concurrent collection?
- 6) What are delegates and built in delegates.
- 7) What is LINQ?
- 8) Single vs SingleOrDefault - First vs FirstOrDefault - Select vs SelectMany
- 9) LINQ Queries practice , left join , right , inner . get 3rd highest
- 10) What is the effect of combining AsParallel with LINQ queries?
- 11) IEnumerable vs toList.
- 12) IQueryable vs toList.
- 13) IQueryable vs IEnumerable.
- 14) What is yeild keyword?
- 15) break vs continue vs return.
- 16) Reflections in c#.
- 17) What is TPL?
- 18) TPL vs Thread
- 19) Async vs Thread
- 20) How to start a Task?
- 21) Does Async create threads ?
- 22) what is async and await?
- 23) difference between Async/Await and TPL.
- 24) how async await work under the hood?
- 25) difference between ValueTask vs Task
- 26) what is ThreadPool Starvation?
- 27) ISyncStateMachine
- 28) What is structure of query pipeline in EF ?
- 29) what latest .net version you have used and whats is special feature you have seen?



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- 30) When we create a new .NET 6 or .NET 7 Console Application, there is no explicit Main method in the Program.cs file. Why is that, and how does the application know where to start execution?"
- 31) What is a concurrency issue in an application, and how do you handle it? Is this related to performance?
- 32) How to do memory optimization ?
- 33) Suppose the database layer is already optimized, but performance issues still exist on the application side. What steps would you take to improve performance at the application level?
- 34) If an application has heavy database interactions, how can we optimize performance—whether it's a legacy application, a modern Web API, or a front-end app like React?
- 35) Imagine there's a production issue with very limited information available. How would you approach diagnosing and resolving the root cause of the issue?
- 36) Which code will perform better ?

```
string longString = "";
for (int i = 0; i < 1000; i++)
{
    longString += "one more ";
}

//-----

StringBuilder sBuilder = new StringBuilder();
for (int i = 0; i < 1000; i++)
{
    sBuilder.Append("Add This");
}
string str = sBuilder.ToString();
```

JAVASCRIPT

THEORY QUESTIONS

- 1) difference in reassign and redeclaration
- 2) var vs let vs const
- 3) types of object creation in js
- 4) what is hoisting concept in js?



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- 5) what do you mean by temporal dead zone?
- 6) what is closure in js?
- 7) what is a prototype?
- 8) what is spread operator?
- 9) what is rest operator?
- 10) Difference between spread and rest operator
- 11) splice vs slice
- 12) What is arrow function?
- 13) what is anonymous function?
- 14) What is array and its types?
- 15) How to flatten multi-dimensional array?
- 16) = vs == vs === state the name of operators and explain the use

ANGULAR

THEORY QUESTIONS -

- 1) Why Angular?
- 2) Which is the latest version going on?
- 3) Difference JS and Angular?
- 4) How to install angular?
- 5) What is AOT?
- 6) What are decorators?
- 7) How do you create a component?
- 8) How does angular app starts?
- 9) Package.json vs package.lock.json file?
- 10) How to download or install other packages?
- 11) What is angular.json file?
- 12) What are modules?
- 13) What is ngModel?
- 14) What is routing and how it is implemented?
- 15) Lazy loading in routing?
- 16) What are directives and its type. Explain with real examples
- 17) How to create custom Directives?
- 18) Types of data binding in angular with examples
- 19) What is string interpolation?
- 20) Difference between string interpolation and Property binding
- 21) What is Http Interceptor? How to implement it?
- 22) What are services in Angular?
- 23) Dependency injection in Angular



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- 24) Ways to pass data from one component to another?
- 25) List down all Life cycle hooks in sequence with examples
- 26) Constructor vs ngOnInit
- 27) ngOnChanges vs ngDoCheck
- 28) What is ViewChild?
- 29) How to create api calls in Angular?
- 30) What are HTTP filters in Angular?
- 31) How to add http headers or authorization in post call?
- 32) What are Pipes and its type?
- 33) How to create Pipes?
- 34) Promise vs Observables
- 35) Observables vs Subject
- 36) Different Types of Subject
- 37) Can we do asynchronous programming in Angular? How?
- 38) How to implement asynchronous programming in Angular?
- 39) Template driven forms and Reactive forms with their difference
- 40) What is auth guard in angular?
- 41) Types of auth guards in angular.
- 42) What is local storage?
- 43) Have you used RXJS? why it is used?
- 44) Give some operators of RXJS
- 45) What is map operator?
- 46) What is Template Reference Variable?
- 47) How to do testing in Angular?
- 48) State management in Angular
- 49) Angular new features

.NET CORE WEB API

- 1) Framework vs Core
- 2) Advantages/features of .net core
- 3) Disadvantages of web api
- 4) How does .net core starts?
- 5) What is middleware in dot net core
- 6) Minimal API
- 7) How did you create a customer middle ware ?
- 8) what is request delegate?
- 9) app.Use vs app.Run



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- 10) What are filters in dot net core?
- 11) What is Authorization filter?
- 12) What is Exception filter?
- 13) What is Action filter?
- 14) Middleware vs filter?
- 15) Configure vs configure Services?
- 16) Dot net core 5.0 vs 6.0
- 17) Can we implement startup.cs file in 6.0?
- 18) What is dependency injection and types
- 19) What is service life cycle? (addTransient, addSingleton, addScoped), state their difference
- 20) What is Routing and its types
- 21) What is an Attribute?
- 22) Serialization and Deserialization
- 23) How to implement Global exception?
- 24) What is session management?
- 25) What is caching and type of caching and how to implement it in dot net core?
- 26) What is MVC pattern?
- 27) What is MVVP pattern?
- 28) What is CORS policy?
- 29) Why CORS no issue in postman but in browser.
- 30) What is Restful API?
- 31) SOAP vs REST
- 32) What is JSON?
- 33) JSON vs XML
- 34) POST vs PUT (Do not use records , idempotent)
- 35) If We can modify resource via Post call then why we need PUT call?
- 36) Can GET call have request body?
- 37) Can we add resource using GET call?
- 38) What is Content negotiations in Web API ?
- 39) If REST API is slow what will you do , where will you hunt , how to optimize ?
- 40) What is Api Versioning and its type? How to implement it
- 41) Return types in .net core?
- 42) Can we have multiple return types in net core?
- 43) How to implement sorting and paging in .net core?
- 44) How to implement Authentication/Authorization in dot net core?
- 45) What is JWT? State its parts. How it is implemented?
- 46) JWT logout.
- 47) When should we use Openid , Oauth as compared to Custom JWT ?
- 48) How to save sensitive data in database?
- 49) What are some common encryption techniques?
- 50) What data encryption techniques you have used in project?
- 51) What is API gateway?
- 52) What is Rate limiter?



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- 53) Suppose There is a requirement to upload a file with lakhs of records in database from mvc.
- 54) How to do it so transaction will be faster
- 55) Can you explain the MVC lifecycle and describe the flow of a request from the browser to the server and back?
- 56) What is cache management? How can it be implemented in a React front-end application? How do we manage caching in an ASP.NET MVC web application? What about in an ASP.NET Core Web API?

ORM

THEORY QUESTIONS

- 1) What is ORM?
- 2) Have you used any ORM in your project ? Which one?
- 3) What is Dapper?
- 4) What is EF core?
- 5) EF core vs framework?
- 6) What are navigation property in EF core?
- 7) How did you implement EF core with relational database?
- 8) Which approach did you follow in your project?
- 9) Code first vs Database first?
- 10) Commands used for code first
- 11) What are fluent APIs?
- 12) How do you call a Stored Procedure from EF core?
- 13) What is lazy loading
- 14) Eager loading in EF core
- 15) 3.Tracking and Non Tracking in EF core
- 16) 4.Can we handle transactions in EF core?
- 17) 5.What is explicit transaction?

Azure

- 1. What is Azure App Service and how do you deploy a .NET application to it?
- 2. What are options to deploy a Web API on Azure? (App Service / AKS / Azure Container Service)
- 3. Why did you choose App Service to deploy WebAPI?
- 4. What is the difference between Azure App Service and Azure VM?
- 5. What is Azure Functions? How is it used with .NET?
- 6. How will you secure Azure Function ?



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7. I want to create storage account and function app. I don't want to enable storage account but Azure Function should read data otherwise hacker can tamper storage account.
8. What are durable function apps ?
9. Check function app is healthy ?
10. 14. I want to receive an email if the function app is failing how do you configure this?
11. What is Azure Key Vault and why is it important?
12. What is Managed Identity in Azure?
13. what is key vault and how to connect kayvault
14. How do you implement CI/CD for a .NET project in Azure?
15. What is Azure Service Bus and how does it work with .NET?
16. What is azure Service bus and how we decide we required queue or topic inside.
17. what is event grid in azure, when we use and why?
18. what is Eventhub xplain with example?
19. What is Data mart ?
20. How do you implement Logging and Monitoring in Azure for .NET applications?
21. Azure APIM Questions
22. How do you secure an ASP.NET Core API in Azure?
23. Explain some AI services in Azure ?
24. Have you used AKS ?
25. How is app insights configured in your ecosystem? Can you see the app insight logs of all of your applications in single place or do I have to go to different URL/Tabs for each of the applications? How do you configure the same?
26. How do you configure the deployment parameters (number of pods/max pods/min pods) for your Web App/API to be deployed to App Service? Do I need to have separate configuration for each of the WebApp/API or can I configure all the apps from single configuration?

SQL

THEORY QUESTIONS

- 1) What is Normalization and state its form?



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- 2) What is BOYCE CODD normal form?
- 3) Indexes in SQL
- 4) Cluster vs Non cluster
- 5) What is execution plan?
- 6) Primary vs Unique
- 7) Union vs Union ALL
- 8) ISNULL vs Coalesce
- 9) Truncate vs Delete
- 10) Table vs Views
- 11) Stored Procedure vs Function
- 12) What is CTE?
- 13) Temporary table vs CTE
- 14) Temporary table vs Temporary Table variable
- 15) Sub Queries vs CTE
- 16) SQL Injection and how will you prevent it?
- 17) Triggers in sql, purpose?
- 18) What is Rownumber?
- 19) Rank vs DenseRank
- 20) What are Magic tables?
- 21) What is Deadlock?
- 22) What is WITH (NOLOCK)?
- 23) What is isolation level?
- 24) What is transaction and How to implement it?
- 25) Merged Statement
- 26) Send unlimited parameters through stored procedures
- 27) in sql How we optimize queries returning 100 millions of records?

Design Pattern and Architecture

- 1) Sometimes they tell to write code also , Singleton pattern code so try to attempt
- 2) What is Design Pattern and why do we need?
- 3) Types of Design Pattern - Creational, Structural, Behavioral
- 4) Which Design patterns you have used in in your project?
- 5) Which Design patterns you know?
- 6) What is Singleton DP? Type down the singleton pattern



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- 7) Connection pooling many answer in singleton but that can be problematic. Especially from DB perspective.
- 8) Singleton pattern disadvantage
- 9) Breaking the Singleton using direct references. Iterator , IEnumerable
- 10) What is Strategy DP?
- 11) What is Adapter DP?
- 12) What is Abstract Factory DP?
- 13) What is Repository DP? list its advantages
- 14) What is CQRS DP?
- 15) What are 3 important elements
- 16) Why is mediator in CQRS
- 17) Why CQRS is important in MS.
- 18) What is event sourcing in CQRS
- 19) Is SQL server Event source ?
- 20) Have you worked on Clean Architecture?
 - 1) What are Microservices?
 - 2) Monolithic vs Microservices
 - 3) Layer vs Tiered
 - 4) Tier and N tier difference
 - 5) How does one microservice call other microservice internally (Queues , WebAPI)
 - 6) Scenario in we have microservice in 2 services, we want to use Service 1 data in service 2 how we can use in between? if we use web Api then how we authenticate in microservices?
 - 7) Strategic DDD vs Tactical DDD.
 - 8) What's a domain and sub domain ?
 - 9) Types of sub domain
 - 10) Bounded context , context map and UL.
 - 11) What is a SAGA pattern?
 - 12) What is DDD?
 - 13) What are different classes , Entity , Value objects and Services
 - 14) Aggregate root.
 - 15) CQRS Pattern
 - 16) Repository
 - 17) Queues.
 - 18) Draw infrastructure diagram
 - 19) How do you decide between a GraphQL and the Backend for Frontend (BFF) pattern?
 - 20) How do you optimize costs in an Azure Cloud microservices architecture?
 - 21) How do you implement the Rate Limiting pattern to isolate resource failures?
 - 22) What is the role of Consumer-Driven Contracts (CDC) in preventing breaking changes?
 - 23) How do you handle "Reference Data" (e.g., countries, currencies): replicate or query?
 - 24) When is the Anti-Corruption Layer (ACL) pattern essential?
 - 25) When do you need a Distributed Lock vs. Optimistic Concurrency?
 - 26) What specific challenges does Event Sourcing introduce in a microservices environment?
 - 27) How do you manage schema versioning for events in an Event-Driven Architecture?



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- 28) When is CQRS (Command Query Responsibility Segregation) necessary versus over-engineering?
- 29) How do you securely inject secrets into services at runtime?
- 30) What strategies do you use for Distributed Rate Limiting?
- 31) How do you approach Multi-Tenancy: Silo model vs. Pool model?
- 32) What are the pros and cons of Client-side vs. Server-side Service Discovery?
- 33) How do you handle time synchronization and clock skew across distributed nodes?
- 34) How do you manage long-running processes without holding open HTTP connections?
- 35) What is your strategy for Canary Deployments in a dependent chain of services?
- 36) How do you design a robust strategy for distributed cache invalidation?
- 37) How do you implement Leader Election for singleton background tasks?
- 38) How does the CAP theorem practically influence your choice of data stores per service?
- 39) How does Conway's Law influence your decisions on team structure and service boundaries?
- 40) How do you manage a "Shared Kernel" without creating tight coupling?
- 41) How do you apply Chaos Engineering to validate system recovery?
- 42) How do you detect and solve the "N+1 Problem" in distributed service calls?

SCENARIO BASED

- 1) How will you increase performance in angular application?
- 2) If there is case of bulk upload photos/videos, what will be your approach?
- 3) If you find slowness in angular application how will you optimize it?
- 4) **If an abstract Employee class has abstract method GetManager() and one concrete method GetSalary().**

And Developer class and QA class inherits it.

Now your Client said there is a requirement to add a CEO class which does not have a manager but has salary.

How you will implement this scenario?

Option - You can extend it as it is or you can change the existing code. But which one would be optimal?

- 5) There are 2 different Interfaces A, B and both are having same method name.
Can a Class e.g. class C, implement both of them?
If yes, then how will it use the method from Class A.



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- 6) There is a sealed class and I want to add new methods to it without changing the existing logic.
- 7) Can a class have multiple constructor with different access modifiers also with static constructor, if yes then in what order it will execute?
- 8) If you need to fetch data but parameter/query string is complex then you will use GET or POST call and why?
- 9) How to increase performance in dot net core?
- 10) How to do bulk insert, update in dot net core?
- 11) How to limit the API hits in dot net core?
- 12) How to handle huge payloads that expires in 30 secs?
- 13) In case of complex logic, will you use linq (dot net core) or SQL (database)?
- 14) How to optimize a Stored procedure?
- 15) If we told you to create a new project then by which approach you will choose Code first or Database first and why?
- 16) How to do bulk insert, update in EF core?

API Testing

- 1) Why we need testing?
- 2) What is Unit testing?
- 3) What is Integration testing?
- 4) Difference Unit and Integration?
- 5) Which unit testing framework you used in your project?
- 6) What is the code coverage percentage?
- 7) What is Moq. Have you used it in project?
- 8) How to handle exception in unit testing?

SCENARIO BASED

```
int[] arr = {2,2,5,6,1,3,3,6,6,5,4,5,5}
```



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Sort the array in descending order of frequency of occurrence of each number. One with highest frequency should come first. And if frequency is same highest number should come first.

For example here 5 is occurring 4 times then 5 should come first

Then 6 is 3 times 6 will be next

2 and 3 both are 2 times. Same fre

quency so 3 should come first then 2

Similar is 4 and 1

So out put will be

{5,5,5,5,6,6,6,3,3,2,2,4,1}

Given a array {3,7,2,4,5,1}

Sort the array in ascending order

Output {1,2,3,4,5,7}

{3,7,3,2,1,4,5,5,2,2}

Eliminate duplicate character from array

Output {3,7,2,1,4,5}

Count the occurrence of each character in string

"hello world"

Output

h 1

e 1

l 3

o 2

w 1



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r 1

d 1

Print the maximum occurring character in string

for example here it will be

l 3

Encode a String

Input AACBAEFEEH

Output should be

A2C1B1A1E1F1E2H1

Decode a string

Input A2C3B1A1

Output should be

AACCCBA

These encode and Decode string where asked in capgemini

Write a program to print following pattern

2

1 4

3 6 5

8 7 10 9

12 11 14 13 16



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*

* *

* * *

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* * *

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*

In Hpcl

These were sql queries asked

EmpName	MobileNo
A	8888888888
B	7777777777
C	8999999999
D	8888888888
E	7777777777
F	7777777777
G	9888888888
H	8888888888

Fetch the names of Employee who have same mobile Number

Here Output will be

A

B

C



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D

E

F

H

As they have same mobile number

tblConsumer

Id Name

1 A

2 B

3 C

4 D

5 E

6 F

tblCustomer

OrderId ConsumerId DeliverDate

1 1 2023-04-01

2 1 2023-04-30

3 1 2023-06-30

4 1 2023-08-31

5 2 2023-01-04

6 2 2023-04-30

7 2 2023-08-31

8 2 2023-08-31

9 4 2023-08-01

10 4 2023-08-31

11 3 2023-06-30



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12	3	2023-01-04
13	3	2023-01-04
14	3	2023-04-01
15	3	2023-04-01
16	6	2023-04-01
17	6	2023-04-30
18	6	2023-08-31
19	6	2023-08-31

Output

RefillCount ConsumerCount

0	2
2	2
3	1
5	1

Write a sql query to generate above output that includes count of consumers based on number of refills(orders) they have received between 1 april 2023 and 30 June 2023. Include those with zero refills.

Example here consumer ID 4 and 5 has no orders place between 1 april and 30 June 2023. So for 0 refills there is 2 consumer count.

Consumer ID 2 and 6 have two orders each between 1 april and 30 June 2023. One is on 1 april and other on 30 april for both. So for 2 refills consumer count is 2

Consumer ID 1 has 3 orders between 1 april and 30 June 2023. So for 3 refills there is only 1 consumer count.

There are no consumers who have place 4 orders. So there is no refill count 4 here

Consumer ID 3 has 5 orders between 1 april 2023 and 30 June 2023. So for 5 refills there is 1 consumer count.

This sql query was also asked in hpcl written test.



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Hint they give as use temp table to store consumerId and refillcount(order count) per consumer between 1 april 2023 and 30 June 2023

Then join consumer table with temp table to get refillcount and consumercount

Suppose there is WebForm on that there is textbox to enter pan numver , in textbox while entering PAN Number.

Example XYGPN1122P. How will you do masking and unmasking?

After XYG it should be *** and in Database it should be saved as XYGPN1122P but in text box

When user enters pan number it should be displayed as

XYG***

In one company this was program was to check if two strings are anagrams. Strings are anagram if one string is obtained by rearranging the letters of another

abac , baac are anagrams

rat , tar are anagrams

India , diani are anagrams

But abc and cbcaa are not anagrams

One of the Big multinational company

- 1) How do you convert your current project to Microservices ?
- 2) If you are asked to design a heavy traffic service with concurrent user requests and data should not be corrupt, how would you do that?
- 3) How do you write middleware components in .Net Core?
- 4) If you have to migrate a project from .Net 4.5 to .Net 8.0 then how would you do it?
- 5) Where do you write AddScoped and AddTransient
- 6) What design patterns have you used?
- 7) What is SSL certificate?



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- 8) How is deployment carried out?
- 9) In entity framework how do you connect to database?
- 10) If you are architect, before project getting into development phase, what all questions you will ask customer?
- 11) What logging mechanism have you used?
- 12) Have you worked with queues or events before?

Azure interview questions

Azure Bus

CODING QUESTIONS

fiddlenet

Array

String

For

- 1) Reverse character in a string:-
 - a. Prepend Approach.
 - b. String Buffer
 - c. Swap
- 2) Reverse words in a string
- 3) Find occurrence of each character in a given string
- 4) Finding two numbers in an array that add up to a target
- 5) Check if a String is a Palindrome
- 6) Reverse Array
- 7) Anagram code
- 8) Move 0's to end of the array
- 9) Find the First Non-Repeating Character
- 10) Remove Duplicates from a String
- 11) Swapping two number
- 12) Reverse a number without converting to string
- 13) Extract the surname
- 14) Find the Longest Substring Without Repeating Characters



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- 15) Remove All White Spaces from a String
- 16) Find the Longest Palindromic Substring
- 17) . int[] GivenArray = { 1, 2, 3, 4, 5, 6 }; //Print Output-> //1 2 3 4//5 6 1 2
- 18) Left/Right rotate an array by K places.
- 19) Prime numbers
- 20) Factorial using recursion
- 21) Find max number in Array.
- 22) Fibonacci series - output = 1, 1, 2, 3, 5, 8, 13, 21
- 23) Basic Star pattern examples
- 24) What is the next number: 1, 3, 7, 13----
- 25) Find Duplicate records in table
- 26) Find Nth highest Salary
- 27) Write a function to sort a given integer array `nums` in-place (without the built-in sort function), where the array contains `n` integers that are either 0, 1, and 2 and represent the colors red, white, and blue. Arrange the objects so that same-colored ones are adjacent, in the order red, white, and blue (0, 1, 2).
Input: `nums = [2, 1, 2, 0, 1, 0, 1, 0, 1]`
Output: `[0, 0, 0, 1, 1, 1, 1, 2, 2]`
- 28) Longest Substring Without Repeating Characters
Problem:
Write a function to return the length of the longest substring in a provided string `s` where all characters in the substring are distinct.
Example 1: Input: `s = "eghghhgg"` Output: 3 Explanation: The longest substring without repeating characters is "egh" with a length of 3. Example 2: Input: `s = "substring"` Output: 8 Explanation: The answer is "ubstring" with a length of 8.
- 29) Maximum Sum of a Contiguous Subarray of Size `k`
Problem:
Given an array of integers `nums` and an integer `k`, find the maximum sum of any contiguous subarray of size `k`.
Example 1: Input: `nums = [2, 1, 5, 1, 3, 2]`, `k = 3` Output: 9
Explanation: The subarray with the maximum sum is `[5, 1, 3]` with a sum of 9.
- 30) Valid Parentheses
Problem:
Given an input string `s` consisting solely of the characters '(', ')', '{', '}', '[' and ']', determine whether `s` is a valid string. A string is considered valid if:
Every opening bracket is closed by a matching type.
Brackets are closed in the correct order.
Example 1: Input: `s = "(){}[]"` Output: True
Example 2: Input: `s = "()){}[]"` Output: False



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LT* Mindtree

.NET / C#

- Stack vs Heap
- What is Garbage Collector? How does it work?
- If I don't want to create instance of a class, how to do it (without static/abstract)?
- How to create Dependency Injection?
- Where does injected dependency reside?
- Constructor chaining
- Partial class – what & use
- Can partial class exist in different projects?
- Types of NullReferenceException
- Task vs Thread / Multithreading
- IEnumerable vs IQueryable (inheritance difference)
- Hashtable vs Dictionary / HashSet vs Dictionary
- Ref vs Out
- IDisposable & where to use
- .NET Framework vs .NET Core (developer perspective)
- Async & Await

Entity Framework / ADO.NET

- What is ADO.NET?
- Can we use Stored Procedure with ADO.NET?
- Difference between SQL query & Stored Procedure in ADO.NET
- EF Migrations commands
- AsNoTracking
- EF Code First vs Database First
- Eager loading

SQL



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- Views vs Indexes
- Indexes (types)
- Triggers
- SP vs Function
- What is SP?
- Where vs Having
- CTE
- Functions – can functions have multiple return types?
- Types of joins
- Query: If commission is null return 0
- Get 5th highest salary
- Department with 0 employees
- How to identify missing index?
- SQL optimization techniques
- Keyword to get last inserted record

ASP.NET Core / MVC / API

- Authentication & Authorization
- JWT (working, expiry, content)
- Logging
- Exception handling
- Middlewares
- REST principles
- Routing
- Model validation
- HTTP methods
- 200 vs 201 status code
- Can we use [Post] on Get action?
- Clean code principles (DRY)



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- Return error message to client
- Multiple views / Same view for different controllers
- DropDown performance handling

Azure

- Azure Functions (data extraction from request)
- Azure Key Vault integration with .NET API
- Handling API issue from Cloud (analysis steps)

Architecture / Design

- Design patterns (Factory)
 - SOLID principles
 - Open/Closed principle violation example
 - Caching (why, memory vs Redis)
 - Microservices architecture
-

◆ EPAM

Coding

- Count repeated characters → Alphabet + Number format
- LINQ group & sum
- Letter count in string
- LINQ join 2 tables
- Parallel programming

Concepts

- DI lifetimes (Scoped explanation specifically)
- Authentication & Authorization
- Logging
- Middlewares
- REST
- Routing



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- Async / Await
-

◆ Caelon

.NET / Project

- Current .NET version used
- Latest .NET features
- Performance difference (.NET vs Framework)
- Ref vs Value types
- JWT working & expiry

MVC / API

- Multiple views same data
- AJAX / jQuery
- Error handling
- Display dropdown data efficiently

Coding

- Letter count program
 - LINQ join
-

◆ Accenture (1st Round)

OOPS

- OOPS concepts
- Inheritance & Multiple inheritance
- Types of Polymorphism
- Calling same method from multiple interfaces

C#

- Swap without 3rd variable
- Delegates
- Async programming



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- Multithreading

MVC / API

- DTO
- ViewBag
- MVC lifecycle
- Action return types
- Security in API

SQL

- Design DB
- Constraints
- SP
- Index & Clustered index

Other

- Microservices
- Hosting
- Attributes

◆ **Cognizant**

Project & Process

- Roles & responsibilities
- Task planning
- Agile
- Estimation
- Client interaction
- Helping juniors

.NET / API

- DI & lifetimes
- Async/Await



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- Authentication & Authorization
- API versioning
- Rate limiting
- Validations before controller
- Exception handling

SQL / EF

- Prevent SQL injection
- Stored Procedures
- CTE
- Functions
- Types of joins
- EF parent-child loading
- Handling scoped DB connection expiry

Azure / DevOps

- Azure services used
- CI/CD
- OAuth

◆ Deloitte

Concepts

- SOLID principles
- Dependency Injection
- JWT mechanism
- Authentication & Authorization
- Async programming
- Exception handling
- CQRS

Coding



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- Singleton implementation
- Delegates (Func vs Action)
- Anagram program
- 3rd largest number

Testing

- Unit testing framework
 - NUnit
 - Mocking vs Stubbing
-

◆ GlobalData

Project

- Module worked on
- ORM used
- Querying DB

ASP.NET

- Constructor in API
- app.Use vs app.Run
- Fix slow MVC page
- Validation before controller

SQL

- 5th highest salary
- Department with 0 employees
- Joins
- Optimization

Caching

- Types of caching
- Redis storage location
- In-memory cache configuration



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◆ Verisk Interview

Project / Basics

- Tech stack
- SQL version
- SQL connection with application

SQL

- Optimize query
- Group patients by doctor (today)

Coding

- 3rd largest element using 1 loop

API

- REST API skeleton
- What is REST API
- Parallel request execution

Persistent - Architect interview questions asked to me

- 1) If a client comes to you for their application modernization, what would be your approach?
- 2) Explain flow of authentication in your Web application / Web API
- 3) Explain flow of authentication in general (JWT / OAuth2)
- 4) What are options to deploy a Web API on Azure? (App Service / AKS / Azure Container Service)
- 5) Why did you choose App Service to deploy WebAPI?
- 6) How do you do logging/tracing in your Web Application / WebAPI? How did you design observability/traceability in your Application ecosystem?
- 7) How do you access data / database in your Web API? Dapper / EF Core / ADO.Net? Which one would you choose under what circumstances?



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- 8) How is app insights configured in your ecosystem? Can you see the app insight logs of all of your applications in single place or do I have to go to different URL/Tabs for each of the applications? How do you configure the same?
- 9) How do you configure the deployment parameters (number of pods/max pods/min pods) for your Web App/API to be deployed to App Service? Do I need to have separate configuration for each of the WebApp/API or can I configure all the apps from single configuration?
- 10) Your application is built in a legacy version of Angular. How would you estimate/approach for upgrading it to the latest framework version?
- 11) How do you conduct TDD of DB objects (Stored procedures)?
- 12) Design patterns used in Event Driven Architecture
- 13) What's use of Key Vault in your project?

5 years experience C**ge**ni

Was asked Mostly Fundamentals was asked

1. Explain D from SOLID with written example
2. Any design pattern which you have implemented. I explained Singleton.
3. How to pass data between controllers
4. Automapper
5. Bundles.config
6. How to implement caching
7. Difference between list and dictionary
8. How to implement security
9. JWT token
10. Return type in API
11. How to handle exception on application level
12. SQL query for 2nd highest salary
13. SQL query for deletion of one of the duplicate records
14. How can we optimize query performance



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15. How to create directory in C#
16. Angular interpolation binding
17. Pipes, impure and pure pipes
18. Give duplicates from a list using LINQ
19. SQL Joins
20. select * from Employees where 1=0. What is the output?
21. Abstract class vs Interface

Per***tent Azure Dotnet Role

- ☐ Middleware
- ☐ Exception handling
- ☐ Performance optimization technique in .NET API
- ☐ Explain your project structure
- ☐ How to make your application more scalable and more available
- ☐ Did you take downtime while deployment? How much? If yes then why?
- ☐ How to achieve 0 downtime (not even 0.00001%)
- ☐ What is Azure Service Bus
- ☐ Azure Service Bus vs Event Hub
- ☐ What is Azure Function App
- ☐ How do you decide to use Function App
- ☐ How to implement Azure Entra ID, what configuration do you have to do?
- ☐ How authentication works with Entra ID?

M***sis

1st round

Intro, roles & responsibilities



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Project & tech stack

Polymorphism

How to call base class constructor

DI & life cycles of DI

Throw vs throw ex

Garbage collector

Cache & types of cache

Middle ware

How to create a custom middleware

App.use() vs App.Run()

Query parameters

How do you extract data from http request

How do you do Model validation (DataAnnotation , Regex)

What is Model binding

Asynchronous programming

Task vs thread

Multi threading

What is the use of IDisposable

Microservices architecture

Design patterns used in project

CQRS

API versioning

If a service is failing continuously, how will you handle this situation

How do you make communication between services

How do you call one service from another

Code :



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1. given sentence and asked to reverse letters in a word without changing the order of words in the sentence
2. Count the occurrence of letters in a string

SQL

Indexes & types

How many cluster indexes will be there for table

CTE vs temp table

Window functions

Rank, dense rank

2nd round

Intro, roles responsibilities

Tell about current project, tech stack, what modules you worked on

Open note pad & Write a controller & show how you implemented repository pattern using DI and entity framework

What DI life cycles used in your project

How do you write unit testing for this ?

What frameworks do you use for unit testing

Middle wares & custom middle wares, how you used middle wares in your project?

Exception handling and how you return user friendly messages to front end



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How you implemented Security in your project

Tell how jwt tokens work

Have you used cors and how does it work ?

What kind of requests it'll block?

What deployments used in your project

EP*M Interview

Epam interview questions

Demonstrates good and well-rounded understanding of .NET Core and .NET Framework differences, including runtime, hosting model, and performance implications.

Proficient in Dependency Injection concepts and practical implementation patterns.

Experienced with application logging using the ILogger interface and follows recommended logging practices.

ASP.NET Core

Shows good understanding of ASP.NET middleware: especially pipeline behavior.

One gap noted: did not correctly articulate the short-circuiting behavior in middleware and instead referenced the circuit breaker pattern.

Good understanding of JWT authentication, token validation, and middleware configuration.

Proficient in model binding and model validation, with clear real-world application examples.

C#

Good understanding of the IDisposable pattern and resource management.

proficiency with async/await and comprehension of Task vs Thread concepts.

Experienced in using Parallel.ForEach and demonstrates very good foundational knowledge of multithreading scenarios.



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Clear understanding of ref vs out semantics and use cases.

Competent in exception handling, including structured use of try-catch-finally.

Successfully implemented a LINQ-based grouping and projection solution to aggregate sales by ProductId, showcasing confidence with collections and object transformations.

Attempted a balanced brackets function with good structure and correct syntax, but was unable to achieve the expected output

LINQ

Demonstrates good proficiency with LINQ.

Skilled in filtering, grouping, aggregation, and projection.

Shows good ability to apply LINQ in real-world business scenarios.

Software Design

Good understanding of DRY, clean coding guidelines, and maintainable design practices.

Capable of creating technical documentation, including coding standards and diagrams.

Design Patterns

Knowledgeable about the Decorator pattern with practical examples.

Understands Singleton and Factory patterns at a conceptual level, though examples were more high-level than deeply implementation-focused.

Entity Framework

Good working knowledge of Code First approach and Code First Migrations.

Understands EF's tracking vs no-tracking queries and how they impact performance and memory usage.

Comfortable discussing EF architecture and real-time usage scenarios.

SQL

Strong understanding of WHERE vs HAVING and SQL query execution order.



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Knows how SQL handles NULL values, including comparisons and function behavior.

Demonstrated logical reasoning by structuring an UPDATE query using a CASE expression, even without recalling exact syntax showing strong conceptual SQL thinking.

Unit & Integration Testing

Demonstrates clear understanding of the AAA (Arrange, Act, Assert) pattern.

Good knowledge of mocking frameworks and how to isolate dependencies.

Can clearly differentiate between unit tests and integration tests, with examples of when each is appropriate.

Microsoft Azure

Shows good knowledge of Azure Functions, serverless execution, triggers, and bindings.

Familiar with Azure Key Vault, including secrets management.

Possesses working knowledge of Application Insights for observability.

Understands fundamentals of Azure Application Gateway

Function apps vs App Services

Authentication Approach - AD

What is JWT

OAuth 2.0 vs Open ID

While doing the API Authentication, Client Credential Flow

Type of Identities, System based Identify, Managed Identify and User Identify

I have a .NET Web API I want to consume SQL Server and I want to Consume Blob Storage, In Azure how the integration looks like

I have a background job which is recurring, what are the different ways to Do, for daily notifications



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Web Jobs

Can I do 2 webjobs in same app Service Plan?

How will you decide I need a separate App Service Plan, looking at scalability or availability

How will you track the issues when it raised in Functions, app Services or etc.. - About Application Insights

Have you worked on Azure Monitor - to track the issues which are attached to Application Insights

Publisher/Subscriber Pattern, How will you identify or track of where is my application, what is my end point and how will you approach.

Did you work on Queues like Service Bus or Storage account queues - Explain me the scenario

I am using the single Topic in SB and multiple subscriber, I want to trigger based on a Field as the parameters in Topics are same how would I achieve that

If you read the queue, how should we process at code level, how ServiceBusQueue class works, DeadLetter Queue or some other concepts in SB works at Code level, explain?

On how many ways you can trigger the Service Bus explain

From .NET I wanted to access the KeyVault - How do I implement it. Explain Class Level

AKS: Have you deploy yourself or DevOps Team, How did you do, Helm Files, Ingress Configuration, Configuration Service Files, CI/CD, etc..

ACR - what is use of it and explain - Mostly used for storing the docker image



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Container and Pod - Difference

Helm Charts - What do Helm Chart do. Why do we need

What CLI you use to test HelmChart in Local machine - Is it with Doctor or something else

Any Experience on Creating VNET or any Networking Concepts

Dependency Injection - Types.. Scoped, Single and Transient

How will you handle the CORS Issues

API End point is consumed by Multiple Applications, How do you manage (Question is related to APIM)

I don't want to write code but I want to validate each request - how can I do in APIM

Did you work on Middleware - Explain me the scenario

How Middleware Executes in a scenario of API Execution to check the Authentication and Authorisation before hitting the Web API

Difference between Sync and Async - give example scenario



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Can we use Async without await ?

I have my Web API I wanted to call an external API - wait once the things done and I wanted to put in a queue base don the response for the further process, what should I Do?

What is the concept of concurrent list and normal list

Different between .Result vs ConfigureAwait

If want to connect to SQL what are all the possible approaches are there. ADO.NET, ORM Frameworks etc..

How do we scale the database when the application is scaling

How will you ensure Connection Pooling

While doing the load testing we simulate the Pooling what type of issues we will face mostly

Have you use LINQ

IEnumerable or IQueryable - what is the difference between and when to use

SQL Server - what did you work and explain.



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Scaler Functions - explain the scenario

How will you check the query taking time - Execution Plan or SQL Profiler

What is Common Table Expression - explain the scenario

For Unit Testing what Tools you used for API's - NUnit, XUnit, and MSTest

GitHubCopilot - any Specific things to define

CQRS and Mediator Design patterns - Explain

Web Page is Unavailable

Vir***a Interview Process (Lead Software Engineer, YOE 10+)

in each L1, L2 and client round They gave multiple problems to solve (provided IDE for both SQL and C#)

In L1, they expect the exact output, so you need to write a complete and correct program.

In L2 and the client round, pseudocode and clear logic explanation are sufficient

1. AI Interview (MCQ Round)

- * Online assessment consisting of multiple-choice questions
- * Covers programming basics, and core concepts

2. L1 – Technical Interview

- * Focus on coding, problem-solving, and fundamental concepts
- * Includes SQL queries, C# programming, and output-based questions



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3. L2 – Techno-Managerial Interview

- * Combination of technical knowledge and practical experience
- * Questions on system design, real-time scenarios, and project discussions
- * May include decision-making and problem-handling approaches

4. Client Round (Technical Test – HackerRank)

- * Hands-on coding test conducted on HackerRank
- * Focus on Communication, problem-solving, coding efficiency, and logic building

5. HR Discussion

- * Final round covering salary, availability, and cultural fit
- * General discussion about background, career goals, and expectations

SQL QUESTIONS

L1

1. Given the tables:

- * Employees
- * Projects
- * EmployeeProjects



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Write a query to:

- * Find employees who have worked on at least 2 projects
- * Identify the longest duration project
- * Return employee names along with the longest project they worked on
- * If multiple longest projects exist, return any one

L2

2. Tables:

- * tblCustomer (CustId, CustName)
- * tbleOrderdtls (OrderId, CustId, OrderNumber)

Write a query to find:

- * CustId
- * CustName
- * OrderCount

For customers having more than 3 orders

3. Laptop Sales:

Tables:

- * Laptop (L_Id, Brand)
- * Sales (Sale_Id, L_Id, Quantity)



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Write a query to display:

* Brand

* TotalQuantity sold

Include brands with zero sales

C# QUESTIONS

L1

1. Given a string "ABC", generate all combinations and permutations

Example: ABC, BCA, CAB, AB, BC, AC

2. What will be the output?

```
class A
```

```
{
```

```
public void Display ()
```

```
{
```

```
Console.WriteLine("A");
```

```
}
```

```
}
```

```
class B:A
```

```
{
```



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```
public void Display ()  
{  
    Console.WriteLine("B");  
}  
}
```

```
B b = new A();  
b.Display();
```

```
A a = new B();  
a.Display();
```

Also consider variations using virtual and override

3. Sort the array

Input: {4,0,7,5,2,8,0,0,0}

Output: {0,0,0,0,4,7,5,2,8}

4. Write a program to find duplicate characters in a string

5. What is Concurrent Programming?

6. What is GraphQL and why is it used?

7. Explain CI/CD

8. Differences:



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- * .NET 6 vs .NET 8
- * var vs dynamic
- * Task vs Thread
- * async vs await

9. Explain:

- * static class
- * middleware (ASP.NET)
- * temp table

Can temp tables be used in:

- * functions
- * triggers

10. How to handle exceptions in:

- * C#
- * React
- * SQL

11. REST API vs traditional API

12. How to implement API versioning

13. ASP.NET Filters



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14. Can we create custom middleware? If yes, how?

15. Alternative for useEffect in React

JavaScript / React Output

16. What will be the output?

```
console.log(true + 1)
```

```
console.log(1 + '2')
```

```
console.log('2' + 1)
```

17. What will be the output?

```
let c;
```

```
console.log(a);
```

```
console.log(b);
```

```
console.log(c);
```

```
var a;
```

```
let b;
```

C# L2



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18. Write a method:

```
public static string MergeStrings(string s1, string s2)
```

Requirements:

* If same length → merge alternately

Example: ABC + 123 → A1B2C3

* If S1 longer → append remaining at end

Example: ABCDEFG + 123 → A1B2C3DEFG

* If S2 longer → append remaining at end

Example: ABC + 123456 → A1B2C3456

TC* Consultancy 20 +LPA Package (.NET Full Stack + AI)

Today, I attended an interview with TCS for a .NET Full Stack role, and I wanted to share some of the key 🔑 questions that were asked. It was a great experience with a strong focus 🧐 on real-time scenarios, problem-solving, and practical knowledge.

📄 .NET Core Questions

- 1) Explain Singleton lifetime in Dependency Injection with a real-time example.
- 2) Explain SOLID principles.



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- 3) Difference between Hashtable and Dictionary.
- 4) Primary vs Secondary Constructor in C#.
- 5) Convert.ToString() vs ToString()
- 6) First(), FirstOrDefault(), Single(), SingleOrDefault() differences
- 7) Middleware + write custom middleware
- 8) GET vs POST - which is better and why
- 9) API performance optimization
- 10) Pagination in .NET Core
- 11) Filters in MVC
- 12) IActionResult vs ActionResult
- 13) Why API is slow and how to fix it

AI-Related Questions

- 1) Explain AI integration in your project
- 2) How to call an AI API in .NET
- 3) What is Prompt Engineering
- 4) Design AI-powered chatbot in .NET
- 5) Chatbot giving wrong answers - how to fix



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? Scenario-Based Questions

- 1) You already have authorization applied to an API, but inside the API method, you need to bypass authorization for a specific condition. How will you implement it in real-time?
- 2) You have `int a = 1`. At runtime, you want to assign dynamically like `var e = a`. Which data type would you use and why?
- 3) Your MVC API returns JSON like `{ "id": 1, "name": null, "age": 25 }`. How will you ignore null values in the response?

📄 SQL Scenarios

Query Behavior:

- 1) `SELECT * INTO newtable FROM oldtable WHERE 1 = 0;`
- 2) `SELECT * INTO newtable FROM oldtable WHERE 1 = 1;`

What will happen and where are these used in real-time?

- 3) Can you write pagination in SQL?
- 4) Insert values like Male = 'M', Female = 'F', Other = 'O'. Write SQL query.
- 5) While creating a UserDetails table, set default age = 18. Write SQL query.
- 4) Find the 3rd highest tanker level of water and also find duplicate tanker names.



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💡 Overall, the interview focused heavily on:

- ◆ Real-time scenarios
- ◆ API performance
- ◆ SQL problem-solving
- ◆ Basic AI understanding

It was a great learning experience!

🧑💡 Tip:

Focus on real-time scenarios, API performance, SQL optimization, and system design thinking.

W**ters interview

What's your approach towards designing a microservice pattern basically? Like, how do you approach

to break down your project and the designing part of it? Like, how you design the solutions?

So, how you define the bounded context, like, if there are overlaps, then how will you take care of this? Okay.

So, if there are overlaps, so, we will have a database, okay, a common database, okay, to take care of that, right, like,

that we call the event sourcing, correct, eventual consistency. So, we will capture the sequence of events, like, user has done

button click, user has done this, this event is created, order is created. .



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Can you explain, like, from practical standpoint, how you implement saga pattern.

So, basically, saga pattern is to take care of distributed transaction, right. We have one transaction, like, payment, invoice,

whatever it is. And then, it says, basically, commit all or commit nothing, right.

So, if something is broken, we will reverse everything, okay. Yeah, I mean, that's the theory part of it.

From practical standpoint,

can you explain how many services you will create? Will saga be a different service, or it will be embedded into your, any of the microservice,

then how will you compensate it from practical standpoint? Okay.

So, where saga sits, basically, first of all?

Yeah. So, where that logic will be implemented?

Let's say, one service is inter-service communication. We are calling one service to another service, then another service to, from the same service,

we are calling multiple service. So, there, we will implement saga pattern.

So, at every places where the communication happens, you will implement saga? So, who is responsible for

implementing the saga, the consumer or the producer,

or who is responsible? Okay. And can you explain event sourcing? You just momentarily talked about event sourcing. Yeah.

Okay. So, how do you connect your .NET Core Web API to any Azure resource which is protected? Azure resource

which is protected? Protected in the sense? It's private, basically. It's private, not available over the internet.



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let's say, I want to manage my SQL connections efficiently, I mean, your .NET Core Web API app is creating SQL connections whenever there is a request

and there are high volume of traffic that is coming in and due to that, it is instantiating so many SQL connections. Okay.

So, but you know, SQL has its own limitation to open number of, on the number of connections to be open at a time, right? So, how you will optimize

or design this situation? Okay. You understood the problem statement, right? Yeah, I understood. Basically,

we are getting so many traffics on SQL connection, right, SQL server side.

Right. So, basically we have to manage that. Connection pool, we can manage, right? We can set a limit in the connection pool, like this 20, 30, 40,

whatever it is and then whatever the extra request will come, we will reject it.

No, we can't reject, we have to serve every request. Okay. So, there is a, okay, there is a pattern, CQRS pattern, okay.

what's your design strategy to design K8 which is deployed into your VNet, like,

how many node pools you have and how many microservices you have and how you design that to be deployed into the K8.?

One pod can have multiple containers. Can a pod have different containers of different types?

whenever the request comes from outside the world, right, from HTTP, from mobile, whatever it is, it comes to ingress and ingress routes the traffic to the appropriate service.

Do you know how a pod connects to a storage account, Azure storage account, or maybe a key vault?

How it connects?



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for microservices, do you think solid principles are applicable?

Do you know sidecar design pattern? Sidecar? Yeah.

Do you know Docker containers ? how to create a container and all?

So, how do you select an image? We, we pull the images, no? Docker pull images, right? No, no, I mean, I mean, what's,

what's the criteria to select an image? What is the criteria to select the image? I mean, how do you decide for this microservice

this should be the image, for that microservice this should be the image, how do you decide? So, that we will, anyway we will pull,

we will, you know, define in our YAML file, right? We have a YAML file, correct? So, we know this... Yeah, but the base, base image

you need to select anyway, right? Yeah, yeah, yeah. So, the base image, how do you decide this base image should be of this type,

this base image should be of that type for this microservice? Because base image is not common for every microservice, right?

So, we have something called tagging, with the help of tagging we can define, we can identify,

I'm just guessing, I'm not sure. We have tagging in the image, okay

We give a name, with the help of tagging we can give a name, okay, in the Docker. Whenever we create a Docker image, Docker image

tagging hyphen T and then we give a name. With the help, obviously we will give a distinct name for all the services, right? There we can, yeah, tagging is there.

you have integrated Azure OpenAI into your project, so, what was the use case?

Do you know spec-driven development?



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FN(LastLetterAlphabet) 4 years Questions

Explain SOLID principles with real example.

Explain Open/Closed Principle using Strategy pattern.

Does Factory pattern violate OCP?

What is Dependency Inversion Principle and how DI implements it?

Why use constructor dependency injection?

How to design interest calculator using Strategy pattern?

When to use Factory vs Dependency Injection?

How to create concrete implementation of a strategy?

Design a simple banking system (Deposit & Withdraw).

Where should business logic be placed (Controller vs Service vs Domain)?

Why should controller not contain business logic?

How do you handle different account types without if-else?

How do you enforce rules like minimum balance?

What is resource leak?

Why should we use using keyword in C#?

What is SQL injection and how to prevent it?

What is race condition in banking transactions?

Why should we use database transactions?

Show total balance across accounts for each customer.

How to optimize query for latest transaction per account?